

MCMC protocol for the memory reproduction model

Target: 10,000 draws from the joint posterior of (μ, σ, ω) and the dummy ω -prior draws used for Savage–Dickey, without visual chain surgery. Source: Tomic & Bays (2023) working-memory reproductions; implemented by `memoryReproductionMCMC.m`.

9,984	72	83	~10k
Trials (100 subjects)	Orientation bins (μ)	Continuous parameters	Discrete ξ nodes / trial-set

Do not publish findKeepChains plus visual drops

The current driver keeps a subset of 12 chains that agree on σ_1 ($\hat{R} \leq 1.1$, minimum 2 chains), then deletes chain 8 by eye because of μ_6 . That is a dataset-specific patch, not a sampler. It conditions inference on a selected mode and is not reproducible for new data.

What this posterior actually is

Each trial shows 3 or 6 oriented lines, then the participant reproduces a probed target. The generative model is a circular Gaussian (wrap-copy of a normal) on the half-circle $[0, \pi)$, plus a swap process: the reported item need not be the target. Nontargets are ordered by similarity to the target, so ω is a ranked swap probability, not an unlabeled mixture.

IDENTIFIED · CONTINUOUS JOINT TARGET	DO NOT MONITOR IN THE MAIN RUN · TRIAL LATENTS
Monitor and diagnose $\mu[1:72]$, σ_3 , σ_6 , ω_3 (3-simplex), ω_6 (6-simplex), and ω -prior copies for Savage–Dickey. That is the joint posterior researchers should export (~83 dimensions after simplex constraints).	$\xi[t]$ (item $\times$ wrap) and posterior predictives $yP[t]$ are Gibbs auxiliaries. They dominate disk, make $\hat{R}$ unreadable, and are not needed for 10k joint parameter draws. Draw predictives in a second pass from stored (μ, σ, ω) or by rerunning with a short monitor list.

Why JAGS needed hand-holding

Mechanism	What goes wrong	Fix
Default μ inits	JAGS draws each $\mu_j \sim \text{Uniform}(0, \pi)$. A random 72-point map is a far mode; some chains never un-permute while also inferring swaps.	Initialize μ at physical bin locations (tiny jitter). Only σ and ω need overdispersion, and even those should stay in the data-supported range.
$0/\pi$ wrap of μ	$\mu[1]=0$ and $\mu[2] \in [0, \pi/2]$ pin global rotation/reflection, but a location near 0 can still stick at π . That is the μ_6 / chain-8 failure.	Data-informed inits almost prevent this. If it still happens, align each chain's μ_j to the nearer of $\{\mu_j, \text{wrap}\}$ relative to the physical bin, then recompute $\hat{R}$ — do not drop the chain by eye.
Discrete ξ mixture	Each trial mixes over item identity $\times$ three wrap copies. Stan's marginalized mixture mixed poorly; JAGS Gibbs on ξ is the right tool, but induces autocorrelation in μ and ω .	Keep JAGS. Thin for storage (5–10), but stop on effective sample size, not raw iteration count.

Mechanism	What goes wrong	Fix
findKeepChains(σ_1)	Selecting the largest subset with $\hat{R}(\sigma_1) < 1.1$ can keep two chains in a shared wrong wrap of μ while looking “converged” on noise.	Require ≥ 4 chains, split- $\hat{R}$ on the full monitored vector, and never subset on a single parameter.

Recommended default (fixed budget)

These settings replace the current $12 \times (2k \text{ burn-in} + 2k \text{ kept, thin } 5)$ run. They are sized for this dataset: $\sim 10k$ trials, 72 bins, pooled subjects. The stored object is $8 \text{ chains} \times 1,250 \text{ draws} = 10,000$ joint samples, sized for kernel Savage–Dickey on ω . Keep all 8 chains.

Knob	Current MATLAB	Shareable default	Why
Chains	12, then drop	8, keep all	Enough to see wrap modes; not so many that subsetting becomes tempting.
Adaptation	Trinity/JAGS default ($\sim 1k$)	2,000	Slice/Gibbs tuners need a long adapt phase on 72 μ 's plus two σ 's.
Burn-in (post-adapt)	2,000	5,000	Lets ξ -averaged μ settle from data-informed starts.
Thin	5	10	ξ -induced autocorrelation; 10 kept draws ≈ 100 updates.
Kept / chain	2,000 (then subset)	1,250	$8 \times 1,250 = 10,000$ joint draws for Savage–Dickey on ω .
Init μ	JAGS prior draw	Physical bins + $N(0, 0.05^2)$ on the half-circle	Highest-leverage change. $\mu[2]$ must stay in $(0, \pi/2)$.
Init σ	Uniform(0, π)	Uniform(0.2, 0.9)	Posterior on this data is ~ 0.53 (set size 3) and ~ 0.66 (set size 6).
Init ω	JAGS Dirichlet(1)	$\omega_3 \approx (0.85, 0.08, 0.07)$; $\omega_6 \approx (0.55, 0.09 \times 5)$, then jitter	Matches the identified similarity-rank coding of swaps.
Init ξ	JAGS default	Column 1 (target, no wrap) for both set-size slots	Responses were already unwrapped toward the target in parse_1.m.
Monitor	$\mu, \sigma, \xi, yP, \omega, \omega$ -prior	$\mu, \sigma, \omega_3, \omega_6, \omega$ -prior	Cuts I/O by orders of magnitude. ω -prior is required for Savage–Dickey.

Wall-clock is similar to the paper run: $8 \text{ chains} \times (2k \text{ adapt} + 5k \text{ burn-in} + 1,250 \times 10) \approx 19.5k$ updates/chain, versus $12 \times 12k$ today, but without monitoring $\sim 20k$ trial nodes. Trinity has no separate adapt command, so `memoryReproductionMCMC.m` sets burn-in to 7,000 ($\sim 2k$ adapt + $5k$ discard).

Better than a fixed budget: sample until the joint is ready

Fixed iteration counts will fail on a new dataset. Treat 10,000 as a **minimum stored Monte Carlo size** after the following stop rule, not as a substitute for ESS. Larger than a typical μ/σ run because kernel Savage–Dickey needs prior and posterior mass near $\omega = (1, 0, \dots)$.

Stop rule (all must pass)

Check	Threshold
Split- $\hat{R}$ (Vehtari et al., 2021) on every monitored node	< 1.05 (try 1.01; do not go above 1.10)
Bulk ESS, minimum over μ , σ , ω	≥ 400 (1,000 preferred)
Tail ESS, minimum over the same	≥ 400
Stored joint draws after combining chains	$\geq 10,000$
Number of retained chains	All 8 (never fewer than 4)

10,000 stored draws is a Monte Carlo sample from the joint, not ESS = 10,000. The extra draws relative to a 5k μ run are for the ω Savage–Dickey kernel density.

CHUNKED LOOP

1. Adapt 2k, burn 5k, collect 1,250 kept draws/chain (thin 10).
2. Align wraps. Compute split- $\hat{R}$, bulk/tail ESS.
3. If any check fails, append another 250/chain and repeat.
4. Cap at 2,500 kept draws/chain. If still failing, restart lost chains from a good chain's last μ , σ , ω plus jitter — do not subset.
5. Stop at 10k joint draws once diagnostics pass. Do not thin the saved file further for Savage–Dickey.

Automated wrap alignment (replaces looking at μ_6)

After every chunk, before $\hat{R}$: for each chain and each bin j , compare the circular mean of μ_j to the physical bin location s_j on $[0, \pi)$. If the wrapped mean (add or subtract π) is closer to s_j , map that chain's μ_j samples through the same wrap. Then diagnose. This is label alignment, not model selection.

R tooling

In R, `runjags::autorun.jags` already implements extend-until-PSRF. Use `psrf.target = 1.05`, `startburnin = 5000`, `startsample = 1000`, `thin = 10`, `n.chains = 8`, and `monitor = c("mu", "sigma", "omega3", "omega6")`.

After `autorun`, `extend.jags` until stored draws reach 10,000. The checked-in `memoryReproduction.Rmd` cannot be that pipeline: it passes `stim` while JAGS expects `s` and `maxPresented`.

What 10k joint draws are for

MARGINALS

σ and ω intervals on this data are already tight. 10k draws with min ESS ≥ 400 is more than enough for the reported 95% CIs ($\sigma_3 \approx 0.53$, $\sigma_6 \approx 0.66$, $\omega_{3,1} \approx 0.86$).

THE M MAP

The scientific object is the 72-vector. Export the $10k \times 72$ matrix (plus σ , ω) so downstream plots and circular CIs use the same draws. Summarize each bin with a circular mean, not a linear mean.

SWAP BAYES FACTORS

Kernel / Savage–Dickey estimates need prior and posterior draws of ω . This run keeps ω -prior on the same chain and stores 10k draws so the density near $(1, 0, \dots)$ is not limited by a short Monte Carlo sample.

Current vs recommended at a glance

Quantity	Paper MATLAB run	Shareable protocol
Chains started	12	8
Chains used for inference	subset + drop 8	all 8
Adapt + burn-in / chain	~1k + 2k	2k + 5k
Post-burn-in updates / chain	10,000 (thin 5)	12,500 (thin 10)
Stored joint draws	up to 24k, then culled	10,000 ($8 \times 1,250$)
$\hat{R}$ gate	σ_1 only, 1.10, min 2 chains	full vector, split- $\hat{R}$ 1.05, all chains
Human inspection	required (μ_6)	none

Protocol for the wrap-copy JAGS model on Tomic & Bays memory reproductions. Implemented in `memoryReproductionMCMC.m` / `memoryReproductionMCMC_jags.txt`. Von Mises variants have an additional $0/2\pi$ endpoint issue in JAGS scalar samplers and should not be the default sharing path until the wrap-copy sampler is automated.